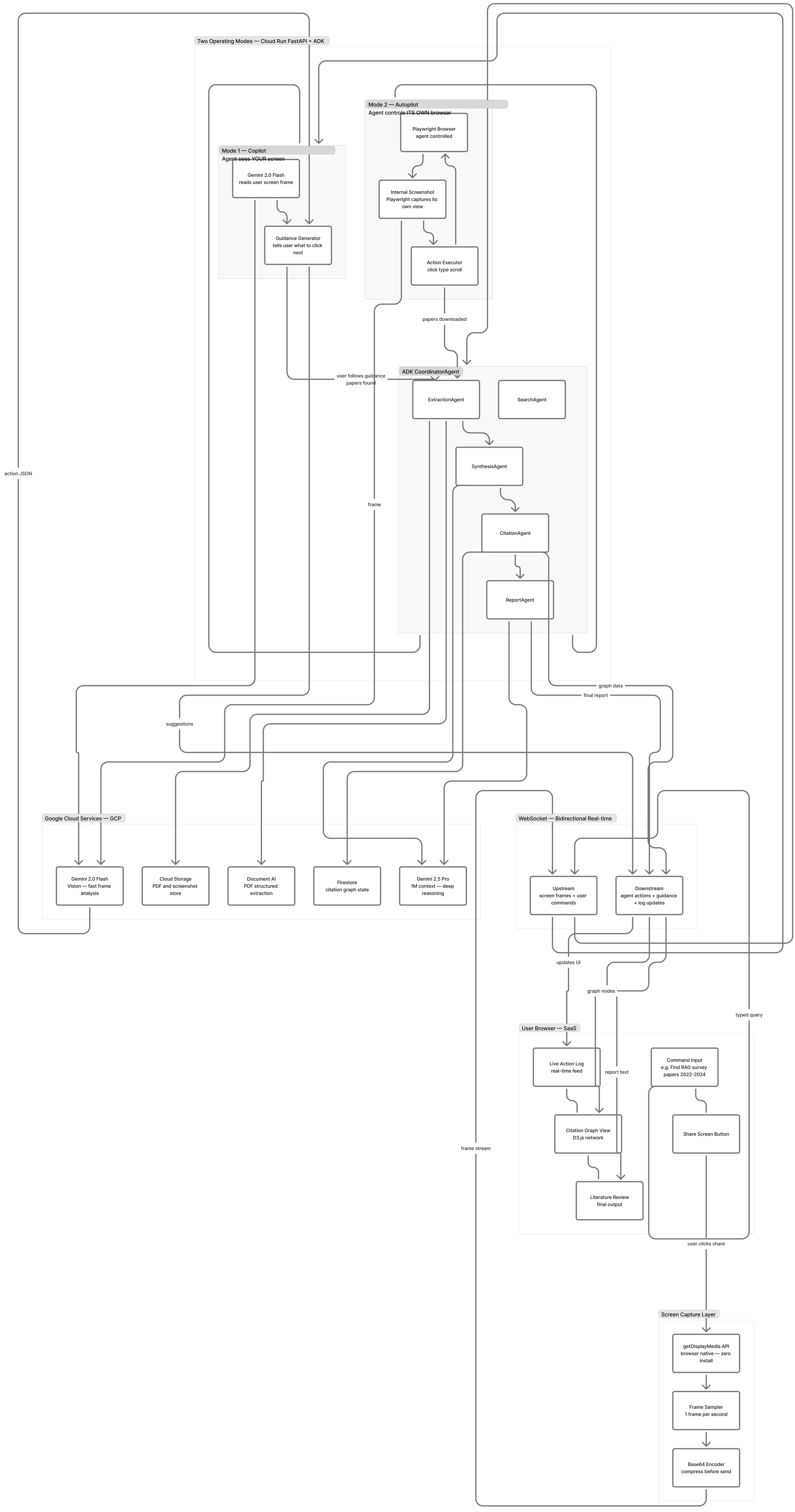
Two Operating Modes — Cloud Run FastAPI + ADK
Mode 1 — Copilot
Agent sees YOUR screen
Gemini 2.0 Flash
reads user screen frame
Guidance Generator
tells user what to click
next
Mode 2 — Autopilot
Agent controls ITS OWN browser
Playwright Browser
agent controlled
Internal Screenshot
Playwright captures its
own view
Action Executor
click type scroll
papers downloaded
ADK CoordinatorAgent
ExtractionAgent
SearchAgent
SynthesisAgent
CitationAgent
ReportAgent
user follows guidance
papers found
action JSON
frame
graph data
final report
suggestions
Google Cloud Services — GCP
Gemini 2.0 Flash
Vision — fast frame
analysis
Cloud Storage
PDF and screenshot
store
Document AI
PDF structured
extraction
Firestore
citation graph state
Gemini 2.5 Pro
1M context — deep
reasoning
WebSocket — Bidirectional Real-time
Upstream
screen frames + user
commands
Downstream
agent actions + guidance
+ log updates
updates UI
graph nodes
typed query
User Browser — SaaS
Live Action Log
real-time feed
report text
Command Input
e.g. Find RAG survey
papers 2022-2024
Citation Graph View
D3.js network
Share Screen Button
Literature Review
final output
frame stream
user clicks share
Screen Capture Layer
getDisplayMedia API
browser native — zero
install
Frame Sampler
1 frame per second
Base64 Encoder
compress before send